

Monthly Programming Contest - July 2025

<https://toph.co/contests/training/dlcxvcm>



Schedule

The contest will run for **4h0m0s**.

Authors

The authors of this contest are shahriar_imran, srfahad2021, and tanvir_005.

Rules

This contest is formatted as per the official rules of ICPC Regional Programming Contests.

You can use Bash 5.2, Brainf*ck, C# Mono 6.0, C++17 GCC 13.2, C++20 Clang 16.0, C++20 GCC 13.2, C++23 GCC 13.2, C17 GCC 13.2, C23 GCC 13.2, Common Lisp SBCL 2.0, D8 11.8, Erlang 22.3, Free Pascal 3.0, Go 1.22, Grep 3.7, Haskell 8.6, Java 1.8, Kotlin 1.9, Kotlin 2.0, Lua 5.4, Node.js 10.16, Perl 5.30, PHP 8.3, PyPy 7.3 (3.10), Python 3.12, Ruby 3.2, Rust 1.84, Swift 5.3, Whitespace, and C11 GCC 13.2 in this contest.

Be fair, be honest. Plagiarism will result in disqualification. Judges' decisions will be final.

Notes

There are 8 challenges in this contest.

Please make sure this booklet contains all of the pages.

If you find any discrepancies between the printed copy and the problem statements in Toph Arena, please rely on the latter.

Disclaimer

The contents of this contest have not been reviewed by Toph and do not necessarily represent Toph's views.

A. Roll Rearrangement

IUBAT assigns each student a unique 8-digit integer ID. Traditionally, in every class, the roll numbers are assigned based on the students' IDs in ascending order. That is, the student with the smallest ID gets roll number 1, the second smallest ID gets roll number 2, and so on.

One day, a new teacher decided to assign roll numbers differently. According to the new rule, roll numbers should be assigned based on the students' brilliance, measured by their CGPA. The student with the highest CGPA should get roll number 1, the next highest CGPA gets roll number 2, and so forth. If two or more students have the same CGPA, the smaller roll will be assigned to the student with the smaller ID.

Given the list of students with their IDs and CGPAs, your task is to determine the roll number of each student according to the new teacher's rule.

Input

The first line contains an integer n ($1 \leq n \leq 2e5$) — the number of students in the class.

The next n lines each contain two values: X and Y

An 8-digit integer X — the unique ID of the i -th student (doesn't contain any leading zeros).

A floating-point number Y ($0.00 \leq Y \leq 4.00$) — the CGPA of the i -th student (exactly 2 floating-point).

It is guaranteed that all IDs are unique.

Output

Print n lines — the i -th line should contain the ID and CGPA of the student having the roll i , separated by a space.

** The CGPA should be printed with exactly two floating point. (eg. 3.00, 3.69 ...)

Example

Input	Output
3 22303266 3.63 22203098 3.92 22203097 3.92	22203097 3.92 22203098 3.92 22303266 3.63
In the test case, the 2nd and 3rd Students' CGPA is same. As the 3rd student has smaller ID, he would get the smaller roll than the second student. The 1st student has	

Input	Output
	lower CGPA than both 2nd and 3rd student, hence he will get a later roll than both 2nd and 3rd student.

B. Bag Battle

Siddique and Chowdhury are brothers. They found a bag on the street. In which, there are n coins, each with a known value. The value of the i -th coin is $a[i]$.

They decided to divide the coins among themselves by playing a game with the following rules:

- The game proceeds in turns; Siddique moves first, then Chowdhury, then Siddique again, and so on.
- On each turn, the current player picks exactly one coin from the bag and keeps it in his pocket.
- The game ends when there are no coins left in the bag.
- Both players want to maximize the total value of the coins they collect.

Assuming both Siddique and Chowdhury play optimally, find the total valuation of coins Siddique and Chowdhury will collect.

Input

The first line contains a single integer n ($1 \leq n \leq 20$) — the number of coins.

The second line contains n integers $a[1], a[2], \dots, a[n]$ ($1 \leq a[i] \leq 10^9$) — the values of the coins.

Output

Print two space separated integers — the total valuation of coins collected by Siddique and the total valuation of coins collected by Choudhury.

Examples

Input	Output
4 1 1 1 1	2 2
Input	Output
4 1 2 3 4	6 4

C. Pipe Puzzle

You are given a grid of size $n \times m$. Each cell in the grid contains a pipe, and each pipe has one or more openings that allow water to pass through. Water can only flow between two **adjacent** cells (up, down, left, or right) **if both pipes have openings on their shared side**.

The goal is to determine whether water can flow from the **top-left corner** (1, 1) to the **bottom-right corner** (n, m) using the pipes.

Pipe Types

Pipe	Openings	Description
+	All directions (U, D, L, R)	Cross pipe
-	Left and Right	Horizontal pipe
I	Up and Down	Vertical pipe
L	Up and Right	Bottom-right L
T	Down, Left and Right	T-shaped pipe

Water can **only flow** between two adjacent cells if:

- The current pipe has an opening towards the neighbor, and
- The neighbor pipe has an opening back towards the current cell.

Input

The first line contains an integer t ($1 \leq t \leq 1e6$) - the number of test cases.

The next t lines contain two integers n and m ($1 \leq n, m \leq 1000$) — the number of rows and columns in the grid.

Each of the next n lines contains a string of m characters, where each character represents the type of pipe at that cell.

It is guaranteed that $n * m$ over all test cases does not exceed $1e6$.

Output

Print "Yes" if water can flow from cell (1,1) to cell (n,m). Otherwise, print "No".

Examples

Input	Output
1 3 3 +-+ I-I L+T	YES

Input	Output
1 2 2 +- I+	NO

D. Nostalgic Necklace

Long ago, a legendary Bengali singer once said:

“তুমি কার লাগিয়া গাঁথরে সখী বকুল ফুলের মালা”

(“For whom did you string this garland of bokul flowers, dear friend?”)

As IUBAT’s “বকুলতলা” is no more, Tanvir wants to construct a necklace using colored stones instead of “Bokul”.

He has a box containing **n distinct colors of stones**.

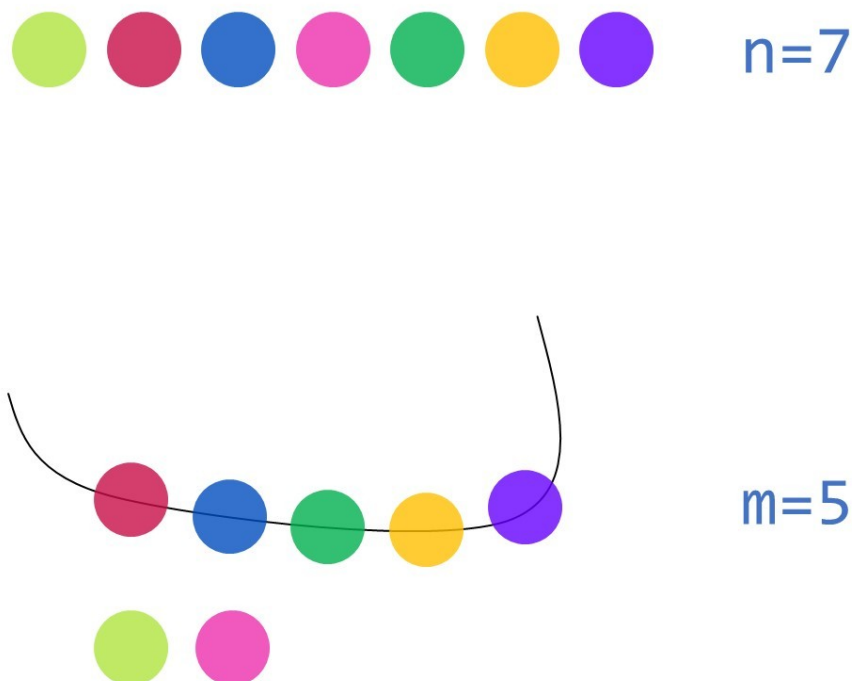
After a long effort, he constructs a necklace with **m stones**, choosing their colors in some particular order.

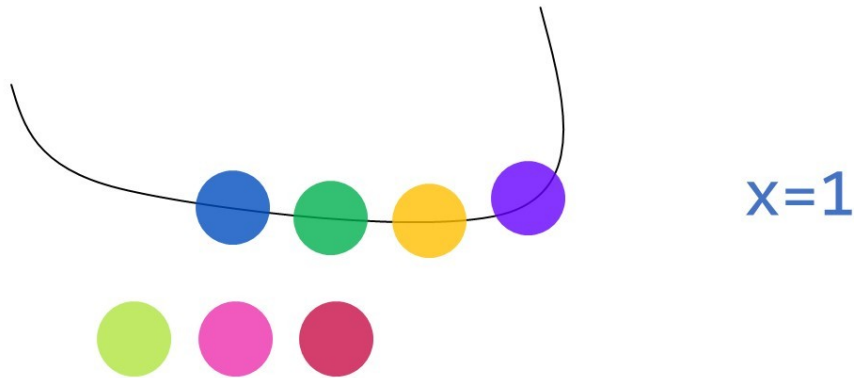
He then puts this necklace untied back into the box along with all the unused stones.

Days later, Tanvir opens the box and is shocked:

x stones from the necklace have fallen off and mixed with the unused stones.

Unfortunately, he no longer remembers the exact color pattern of the previous necklace. He only knows from which position the stones have fallen off.





Now Tanvir wonders:

If he randomly picks stones from the box to fill the empty positions in the necklace, what is the probability that the resulting necklace is exactly same as the previous one (matching every stone's color in the same order)?

You are given **q queries**, each with a different value of x , and you must compute this probability for each.

As we want to avoid floating values and large numbers, print the answer by modulo $1e9 + 7$.

Input

The first line contains three integer numbers n , m , q - the total number of stones, number of stones used to construct necklace and the number of queries.

Next q lines of independent queries containing exactly one integer x - the number of stones that are fallen off from the untied necklace.

Constraints:

$(0 \leq n \leq 10000000, 0 \leq q \leq 100000, n \geq m \geq x)$

Output

Output q lines where each line is the answer for the corresponding query as described in the statement.

Examples

Input	Output
1 1 1 1	1

Input	Output
3 2 1 1	500000004

E. Perfect Partition

Karima found a strange stick of length n in her backyard. It was no ordinary stick, rather it had a puzzle written on it!

The puzzle said:

“This stick can be broken into four parts.

The length of each part must be a perfect square number (like

0, 1, 4, 9, 16, ...).

The summation of the lengths of the four parts must be equal to the full stick length.”

She knows that *any* stick can be broken this way, but she doesn't know how to figure it out. So, she's asking you for help!

Your job is simple:

For each stick Karima has, find **a, b, c, d** — four non-negative integer such that the sum of their square is equal to **n**.

** If there are multiple solutions, you can print any of them.

Input

The first line has an integer t ($1 \leq t \leq 1000$) — the number of test cases.

Each of the next t lines has an integer n ($0 \leq n \leq 1e7$) — the length of the stick.

Output

For each test case, print four space separated non-negative integers

a, b, c and **d** that satisfy :

$$n = a^2 + b^2 + c^2 + d^2.$$

Example

Input	Output
3	1 1 1 1
4	3 2 1 1
15	4 2 0 0
20	

F. Chrono Conundrum

The International University of Business Agriculture and Technology (IUBAT) is the first private university in Bangladesh, established in 1991 by Prof. Dr. M. Alimullah Miyan with a vision to democratize higher education through a model called Knowledge-Based Area Development (KBAD), aiming to produce at least one graduate from every village and region of the country. Located in Uttara, Dhaka, IUBAT operates on a lush, eco-friendly campus by the Turag River and offers a wide range of undergraduate and graduate programs in business administration, computer science, electrical, civil and mechanical engineering, agriculture, nursing, economics, English, and tourism & hospitality management. The university is fully approved by the University Grants Commission (UGC) and is a member of the Association of Commonwealth Universities, with many of its credits transferable to institutions abroad. The holiday for student of IUBAT is Thursday and Friday. IUBAT uses English as the medium of instruction and follows a trimester system, allowing flexible study plans. It is known for its commitment to quality education, affordable tuition, and inclusive student support, including scholarships, interest-free cooperative financing, health insurance, and free transport.

Kamran is a student of IUBAT, studying Computer Science and Engineering. He is a very good and hardworking student. Because he did well in SSC and HSC exams, the university gave him a 100% waiver. So now, he only needs to pay the registration fee each semester.

Yesterday was a very busy and stressful day for Kamran. He had to attend 5 hours of class in IUBAT and felt very tired afterward. When he woke up today and saw that it was an off day, he wanted to relax and do something fun. So, he called his friends and seniors and made a quick plan to visit Mawa for a day.

Kamran and his friends reached Mawa at 12:30 PM. The weather was nice, and the air near the Padma River felt fresh. They took a boat ride, clicked some photos, and ate the famous ilish fish from a small hotel near the river. They laughed, had fun, and talked a lot.

Later in the afternoon, they sat by the river, watching the boats and the Padma Bridge. Kamran felt happy and relaxed. He understood that taking small breaks is important when life gets busy. When the sun started to set, the sky looked beautiful with colors like orange and purple. Kamran looked around and felt thankful — for his studies, his friends, and the happy time he had that day.

They came back home in the evening, tired but with big smiles on their faces. That night, Kamran slept early, feeling fresh and energetic.

Input

In this problem, there is not input.

Output

From the Above story, print what day is tomorrow?

** Print format : First character should be upper case and rest of the character should be lowercase english letters.

G. Royal Route

In the Kingdom of **Infinitas**, a grand parade is about to begin. Along a long 1-dimensional road lie countless white tiles, numbered $0, 1, 2, 3, 4, \dots$ forever to the east.

To prepare for the King's arrival, two skilled workers are sent to paint tiles:

- **Worker A** starts from the beginning and paints every **a-th** tile:
tiles $a, 2a, 3a, 4a, \dots$
- **Worker B** starts from the beginning and paints every **b-th** tile:
tiles $b, 2b, 3b, 4b, \dots$

When the parade begins, the King stands on tile 0 and declares:

"I will not step on tiles that are not painted."

Saying this, the king starts walking to initiate the grand parade. After the k -th step, the king stops and announces the initiation of the grand parade.

You have to determine the number of the tile from which the king initiated the grand parade.

Input

The first line contains an integer $t (1 \leq t \leq 1e5)$ $t \leq$ — the number of test cases you have to answer.

Each of the next t lines contains three positive integers a, b and k ($1 \leq a, b, k \leq 1e9$).

Output

For each testcase, print the answer in a new line.

Example

Input	Output
2 1 1 1 6 9 2	1 9

H. Launch Light

Far away on a quiet planet, there's a hidden space station called **AstroPoint**. It helps spaceships launch safely into space. But instead of using radios or messages, they send a special **arrow-shaped light signal** into the sky.

This signal has three parts:

- A sharp **arrowhead** at the top, pointing upward
- A straight **body** in the middle
- And a **tail** at the bottom, shaped like a upward-pointing arrow

When the astronauts see this full arrow in the sky, they know it's time to begin the launch. Sometimes, depending on the mission, the signal needs to be of different sizes.

Your task is to write a program that creates the arrow signal based on the given size.

Input

The first line contains an integer t — the number of test cases.

Each of the test cases contains an even integer n ($2 \leq n \leq 2000$)

This number tells you how tall the arrowhead, body and tail should be.

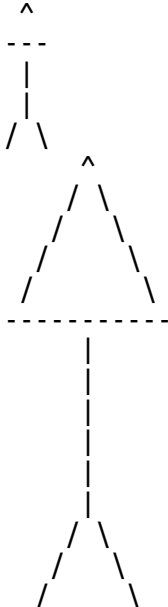
Output

Print the arrow-shaped signal in the form of characters ($\wedge$, $-$, $/$, $\backslash$, $|$) and spaces, structured as:

1. **Top Arrowhead** of height n , centered.
 - The **first line** of the arrowhead should print only $\wedge$
 - The **last line** should print a full line of $-$
 - The **left and right borders** are formed using $/$ and $\backslash$ for the rest of the lines
2. **A straight Body** of height n , aligned in the center, made using $|$
3. **Bottom Arrowhead** of height $n / 2$, pointing upward, also centered.
 - This part uses $/$ and $\backslash$ for the left and right borders
 - It does **not** contain the $\wedge$ symbol like the **Top Arrowhead**

****Please don't print any extra (unnecessary) space or new line.**

Examples

Input	Output
2 2 6	
Input	Output
1 4	